



Western Australian Certificate of Education Examination, 2012

Question/Answer Booklet

PHYSICS

Stage 2

Please place your student identification label in this box

Student Number: In figures

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In words

Time allowed for this paper

Reading time before commencing work: ten minutes
Working time for paper: three hours

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Formulae and Data Booklet

Number of additional
answer booklets used
(if applicable):

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To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction tape/fluid, eraser, ruler, highlighters

Special items: non-programmable calculators approved for use in the WACE examinations,
drawing templates, drawing compass and a protractor

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Short answers	17	17	70	68	40
Section Two: Problem-solving	6	6	90	85	50
Section Three: Comprehension	1	1	20	17	10
Total					100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2012*. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer Booklet.
3. When calculating numerical answers, show your working or reasoning clearly. Give final answers to three significant figures and include appropriate units where applicable.

When estimating numerical answers, show your working or reasoning clearly. Give final answers to a maximum of two significant figures and include appropriate units where applicable.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.
6. The Formulae and Data Booklet is **not** handed in with your Question/Answer Booklet.

Section One: Short answers**40% (68 Marks)**

This section has **seventeen (17)** questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Give final answers to three significant figures and include appropriate units where applicable.

When estimating numerical answers, show your working or reasoning clearly. Give final answers to a maximum of two significant figures and include appropriate units where applicable.

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Suggested working time: 70 minutes.

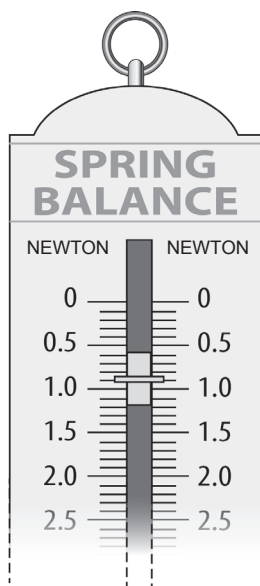
Question 1**(3 marks)**

A hiker is walking east across a flat desert. After walking for 945 m, he realises he has dropped his water bottle and walks back 570 m to get it. Draw and label a vector diagram showing his path and calculate his resultant displacement.

Question 2

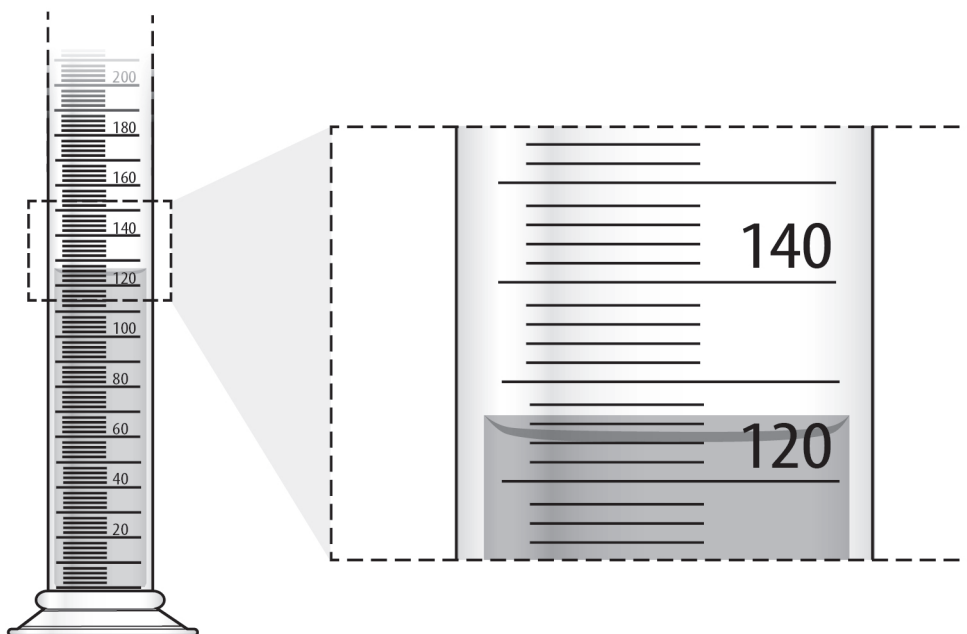
(4 marks)

For the following pictures read the scale, record the measurement and determine the uncertainty of the measurement.



The force measured by the spring balance is _____ N.

The uncertainty is _____ .



The volume of the liquid in the graduated cylinder is _____ mL.

The uncertainty is _____ .

See next page

Question 3**(3 marks)**

Draw and label a diagram to show your understanding of the currently accepted structure of a helium-4 atom.

Question 4**(3 marks)**

A sprinter completes one lap of an oval 400 m track in 55.0 s, finishing at the same point he started.

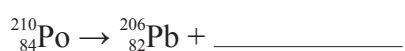
- (a) Determine the speed of the sprinter. (2 marks)

- (b) Determine the velocity of the sprinter. (1 mark)

Question 5**(5 marks)**

The radioactive isotope polonium-210 decays directly to lead-206, which is stable.

- (a) Complete the equation below showing this decay. (2 marks)



- (b) The half-life of polonium-210 is 138 days. If a sample of polonium-210 has an initial activity of 2000 Bq, calculate how long will it take before the sample's activity decreases to 125 Bq. (3 marks)

Question 6**(4 marks)**

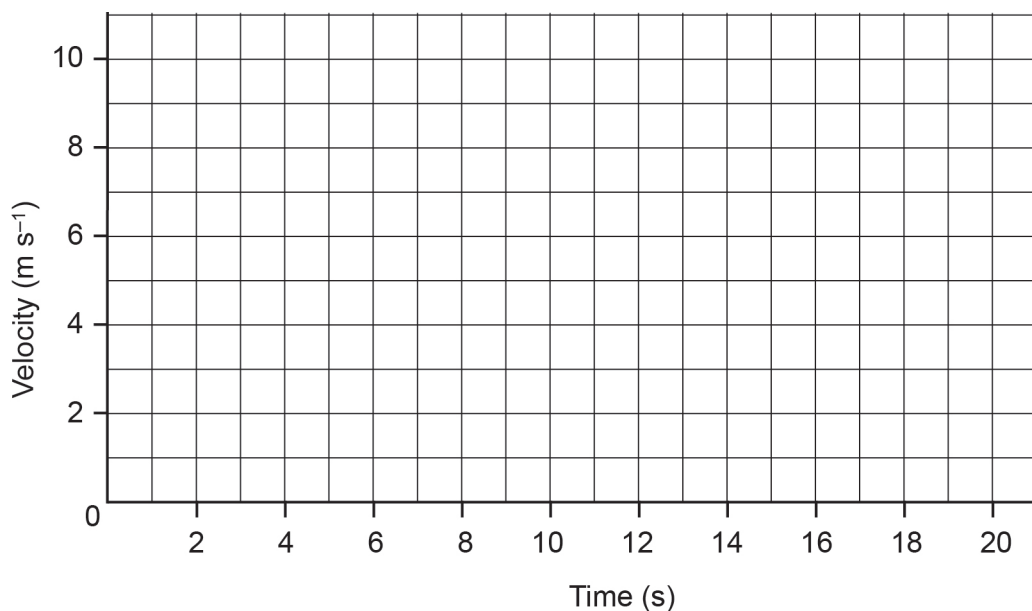
A toaster malfunctions and does not eject the toasted bread. Describe the possible dangers associated with getting the toasted bread out of the toaster with a metal-handled knife.

Question 7

(5 marks)

Jill begins from rest and after 1.0 s is moving at a velocity of 8.0 m s^{-1} for 9.0 s. She then takes 3.0 s to slow down to 2.0 m s^{-1} and continues at this velocity.

- (a) Graph Jill's motion on the velocity-time graph below from the information given above. (2 marks)



- (b) Calculate Jill's acceleration between the 10th and the 13th second. (3 marks)

Question 8

(6 marks)

Complete **two (2)** circuit diagrams, which consist of one 1.5 V cell lighting two light globes.

- (a) Draw one circuit with the light globes in parallel and the other with the light globes in series. (3 marks)

Parallel circuit	Series circuit

- (b) Assuming that all components are the same in each circuit, explain which circuit will have the brighter light globes. (3 marks)

[illegible]

The following information is used for Questions 9 and 10.

Geologists collect samples from active volcanoes as shown in the following pictures. The samples are taken, when the molten rock is about 1150°C , with a metal hammer that has a pointed side. The sample is then cooled quickly by placing it and the hammer into a bucket of water.



Question 9

(4 marks)

Select an article of the geologist's clothing and explain what property it has that would make it useful for the hot environment, and what form of heat transfer it is used to address.

Question 10

(4 marks)

A geologist takes a 1.78 kg rock sample back to a laboratory and uses an oven that can measure the amount of heat transfer in order to make an accurate measurement of the rock's specific heat capacity.

- (a) Calculate the specific heat capacity (including the units) of the rock sample, given the rock absorbs 1.57×10^6 J of energy to increase its temperature by 1050°C . (3 marks)

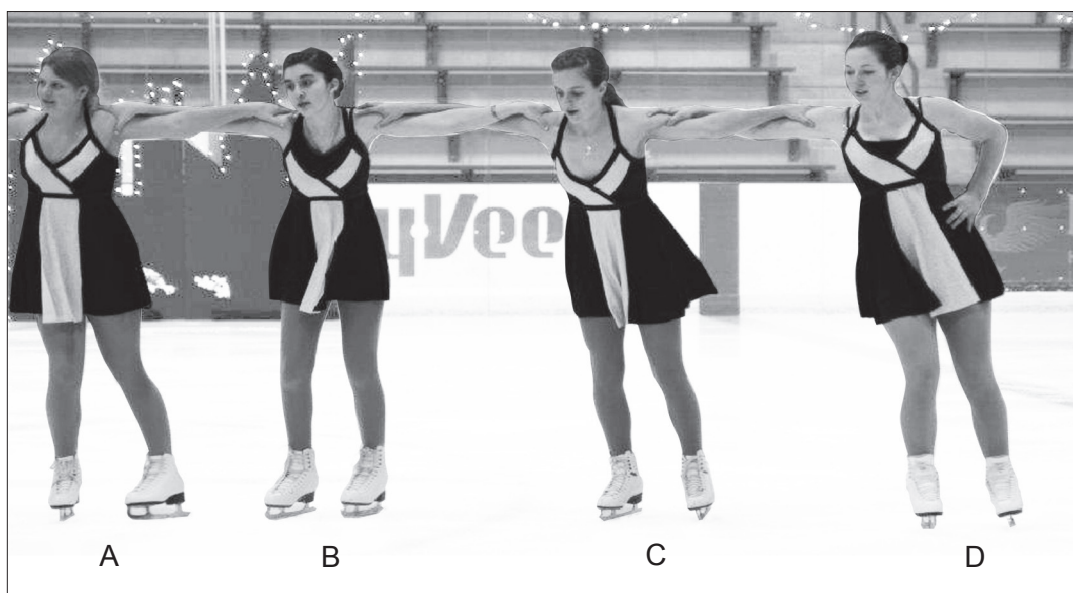
- (b) The geologist could have estimated the specific heat capacity of the rock by measuring the temperature of the water in the bucket before and after adding the hot solidified rock and hammer to it. Give a reason why this method would not give an accurate result.

(1 mark)

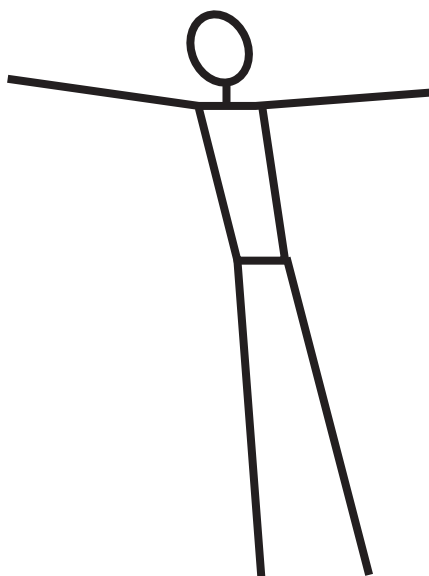
Question 11

(3 marks)

Four ice skaters are skating in a line at constant speed, as shown on the picture below.



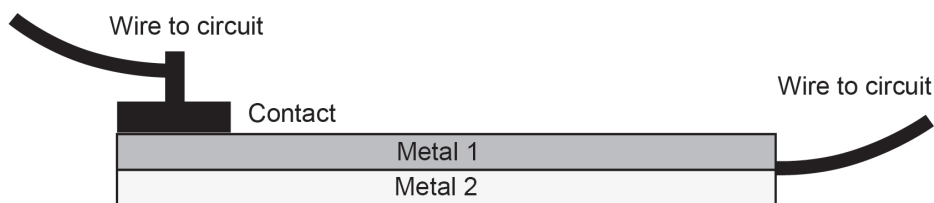
Use the stick figure below to draw a free body diagram which clearly shows the forces acting on skater C.



Question 12

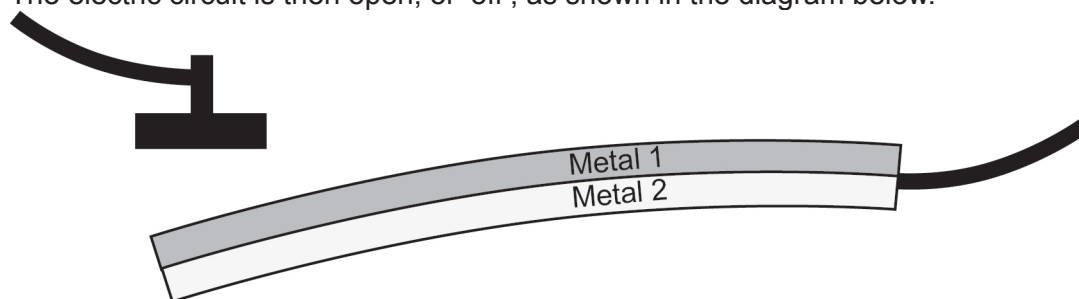
(4 marks)

A common type of switch used for regulating temperature is a bimetallic strip. A bimetallic strip is made of two different types of metal. The bimetallic strip is set so that when a certain temperature is reached, a contact is made and the circuit is closed or 'on' as shown in the diagram below.



Bimetallic strip in the 'on' position

Both metals experience the same change in temperature in an oven. When the temperature is too high, the contact is broken because the different properties of the metals make the strip bend. The electric circuit is then open, or 'off', as shown in the diagram below.



Bimetallic strip in the 'off' position

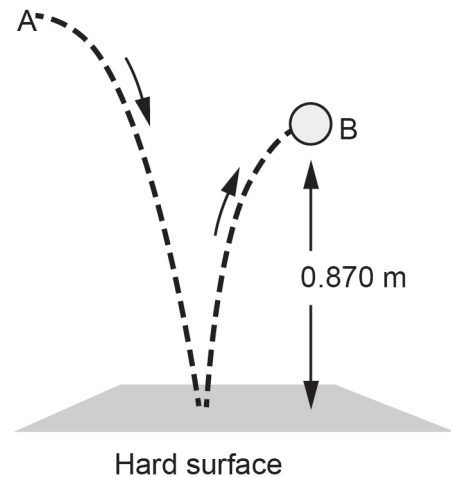
Complete the following table by using the diagram(s) and your understanding of the concepts of thermal expansion and contraction to choose the best answer. Tick the appropriate column to indicate your answer to each question.

Question	Metal 1	Metal 2	Neither
Which metal expands more when heated?			
Which metal has particles with greater average kinetic energy when in the 'off' position?			
Which metal's particles have the least change in their average distance apart when cooled?			
Which metal's particles have the least change in their average distance apart when heated?			

Question 13

(4 marks)

A 0.250 kg ball bounces on a hard surface after being dropped from a height. The ball retains 80% of its kinetic energy in the collision and rises to a maximum height of 0.870 m above the ground. Calculate its potential energy at A.



Question 14

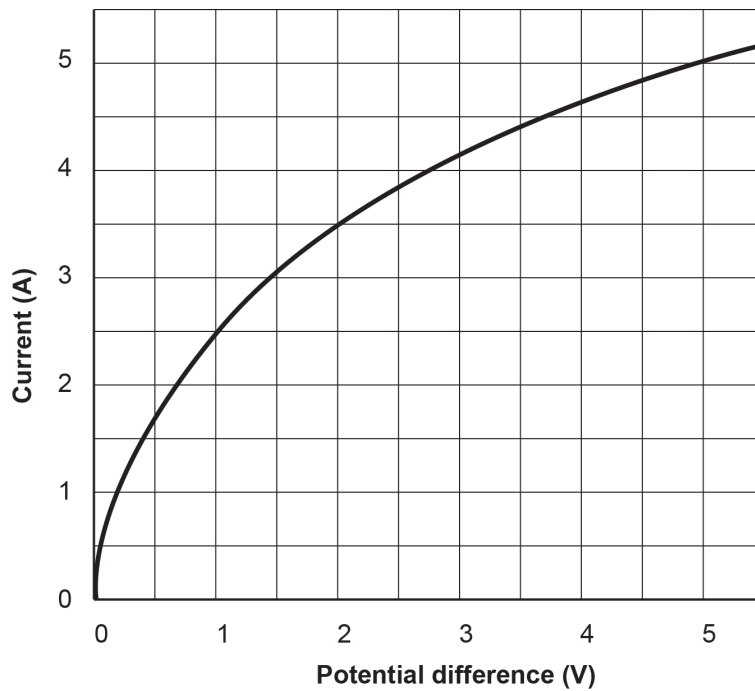
(3 marks)

Annie carries a suitcase into a lift. As the lift starts to move, she feels the suitcase change weight and feel lighter. Explain how she can tell which way the lift is moving.

Question 15

(4 marks)

Chris is measuring the current flowing through a non-ohmic conductor as she varies the potential difference of the power source. The results are graphed below.



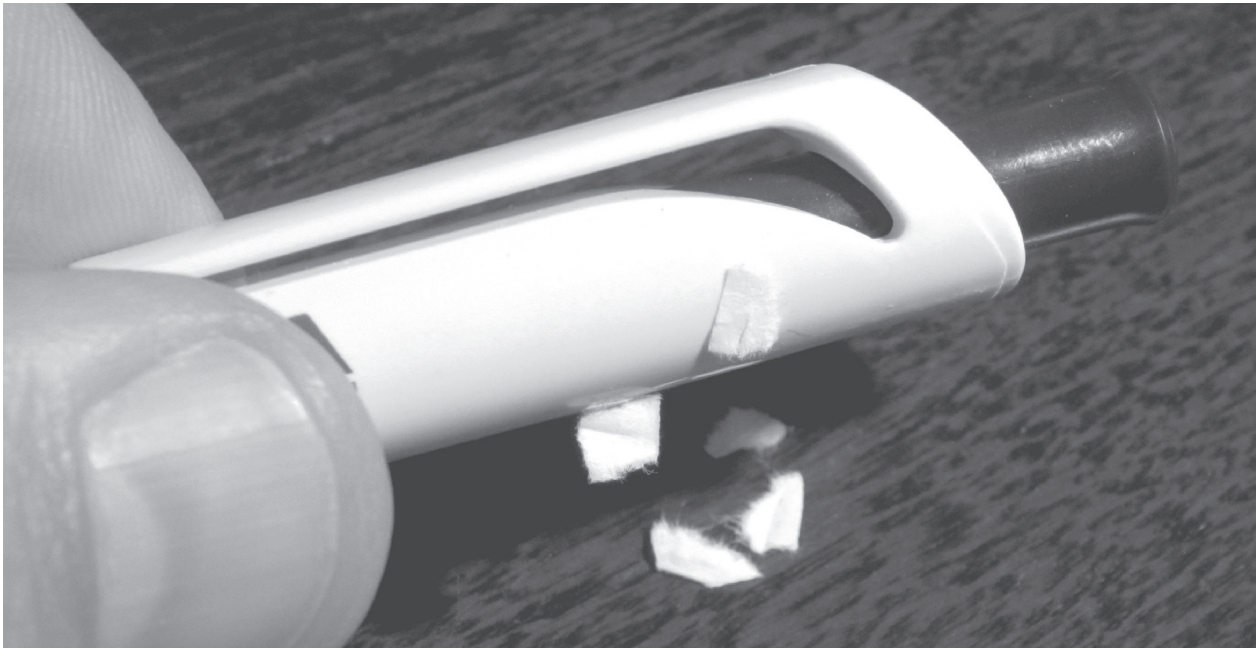
- (a) Explain what is meant by the term 'non-ohmic'. (2 marks)

- (b) Calculate the resistance when the potential difference is 1.0 V. (2 marks)

Question 16

(4 marks)

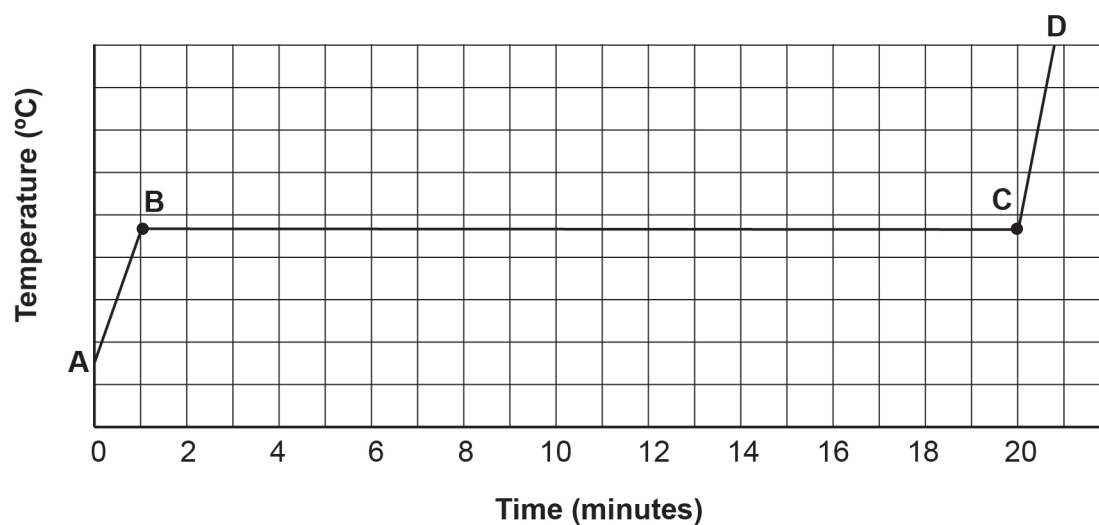
A plastic pen, when rubbed with wool, is then seen to be able to attract small pieces of paper from a desktop. Once the paper touches the pen, the paper flies off. Explain these phenomena.



Question 17

(5 marks)

The following graph shows the temperature change when 1.00 kg of water is heated in an insulated container by a 2.00 kW heater. The temperature change has not been recorded.



Using the information contained in the graph,

- (a) determine the time interval, in seconds, between B and C; and (2 marks)

- (b) use this time value to calculate the amount of energy supplied by the 2.0 kW heater in the time interval between B and C. Express your answer with appropriate units. (3 marks)

End of Section One

See next page

Section Two: Problem Solving**50% (85 Marks)**

This section has **six (6)** questions. Answer **all** questions. Write your answers in the spaces provided.

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Suggested working time: 90 minutes.

Question 18**(16 marks)**

Photographs of three rechargeable batteries/cells are shown below. For each of the following questions, assume they are fully charged.



Photograph A:
AA Rechargeable cell



Photograph C:
Rechargeable camera battery



Photograph B:
**Rechargeable mobile
phone battery**

- (a) The unit mAh is a common unit referring to the amount of stored charge that is available, where 1mAh (one milliamp hour) is equal to 3.6 coulombs. Show by appropriate calculation(s), that 1mAh = 3.6 C. (3 marks)

- (b) Fill in the first two columns of the table for the three power sources using information from the photographs. Then calculate the stored charge and enter the value in the third column. The first row has been completed for you. (4 marks)

Photograph	Potential difference (V)	mAh	Stored charge (C)
A	1.2	2450	8820
B			
C			

- (c) Calculate the work that can be done by the AA rechargeable cell in Photograph A, assuming all of the charge could be released at the given potential difference of 1.2 V. (2 marks)

- (d) Determine whether the following statements are true or false. (3 marks)

	Statement	True or False
(i)	The battery in Photograph C, at 3.7 V has more stored charge than the one at 1.2 V in Photograph A.	
(ii)	The batteries in Photographs B and C at 3.7 V will both light the same type of globe for the same length of time.	
(iii)	A globe will be brighter when connected to the cell in Photograph A than when connected to the battery in Photograph B.	

- (e) The cell in Photograph A completely discharges in one hour (3600 s) when connected to a circuit. Calculate the resistance of the circuit. Assume the potential difference of the cell remains constant during this time. (4 marks)

Question 19

(13 marks)

Jake is working on the top of a building which is 21.5 m above the ground. He drops both a nut and bolt.

- (a) The nut has a mass of 5.55×10^{-2} kg and the bolt has a mass of 2.51×10^{-1} kg. Which of the two will reach the ground first? Explain your reasoning. (2 marks)

- (b) Assuming the bolt falls straight to the ground and there is no wind resistance, calculate

- (i) the time it takes for the bolt to reach the ground; and (3 marks)

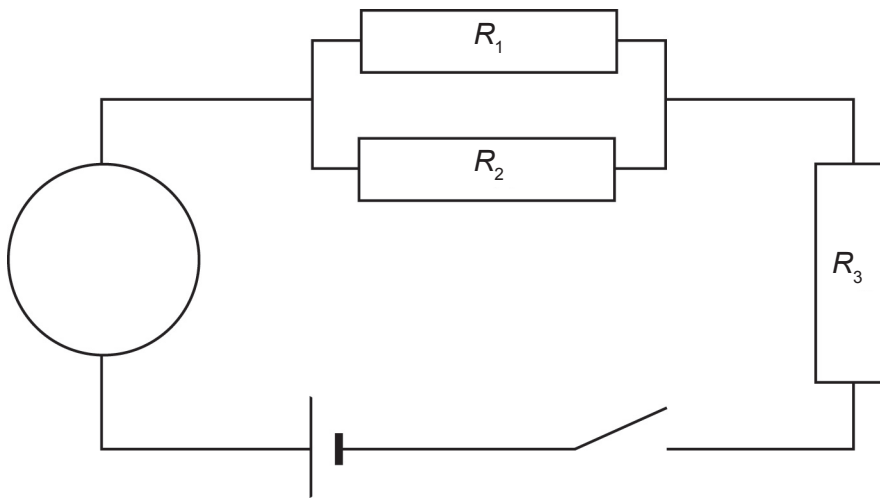
- (ii) the final velocity of the bolt as it hits the ground. (3 marks)

- (c) The ground consists of soft soil and the bolt enters the soil and stops 2.25×10^{-2} m below the surface. Calculate the magnitude of the force that the ground has exerted on the bolt. If you were unable to determine an impact velocity in Part (b), use the value 25.0 m s^{-1} . (5 marks)

Question 20

(14 marks)

An electrical circuit was constructed as shown below. It has a meter, a 1.50 V cell and a switch. The total resistance R_T of the circuit is 9.24Ω .



- (a) The circle in the circuit indicates where a voltmeter or an ammeter should be placed.

In the circle on the diagram above, clearly write the letter 'A' for ammeter or 'V' for voltmeter, indicating which meter should be correctly placed here. Explain why you have made this choice of meter. (3 marks)

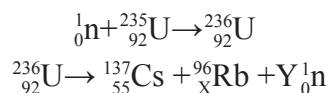
- (b) R_2 has a resistance of $12.0\ \Omega$. Given the total resistance of resistors R_1 and R_2 is $3.00\ \Omega$, calculate the resistance of R_1 . (3 marks)
- (c) The cell provides $1.50\ \text{V}$ of potential difference to the circuit, which has a total resistance, R_T of $9.24\ \Omega$.
- (i) On the diagram on page 22, use an arrow to indicate the direction of conventional current in this circuit.
Show through calculation that the current flowing through R_3 is $0.162\ \text{A}$. (3 marks)
- (ii) Calculate the current flowing through R_2 . (5 marks)

Question 21

(15 marks)

Food that is intended for storage needs to be treated to minimise the presence of organisms that could cause the food to become spoiled and unfit for eating. One method used to prolong storage is irradiation, which is achieved by exposing the food to a radioactive isotope to kill any contaminating organisms. The isotope caesium-137, which has a half-life of 30.2 years, is commonly used for this process.

- (a) Caesium-137 is not available naturally and must be produced through nuclear fission. Neutron bombardment of uranium-235 creates the parent isotope uranium-236 which undergoes fission. One possible fission outcome would create caesium-137, rubidium-96 and some neutrons as shown in the equations below.



Determine the values for X and the number of neutrons Y.

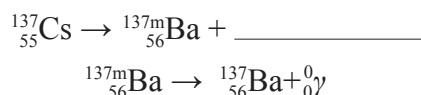
(2 marks)

X = _____

Y = _____

- (b) Caesium-137 undergoes radioactive decay to form a short lived, excited, metastable barium-137m (m for metastable) and another particle. The barium-137m then undergoes gamma emission to become a stable barium isotope.

- (i) Complete the equation below for this decay chain, showing clearly the missing particle. (2 marks)



- (ii) Outline **two (2)** properties that make caesium-137 a good choice for food irradiation. (2 marks)

One: _____

Two: _____

- (c) Caesium-137 has a half-life of 30.2 years, whereas barium-137m has a half-life of 153 s and barium-137 is stable. Using this half-life information, explain which of these two unstable isotopes would be more abundant in a sample used for irradiation in the following situations. (4 marks)

A newly processed sample of caesium-137 ready to be used for irradiation would contain more of the unstable isotope _____ because

A 25 year old sample of caesium-137 ready to be replaced would contain more of the unstable isotope _____ because

- (d) To increase the shelf life of some chicken meat, it is exposed to radiation. How much energy is absorbed by 1.50 kg of meat when given a dose of 3000 Gy? (3 marks)

- (e) Name a precaution used by workers in this situation to limit their exposure to ionising radiation and explain how it reduces exposure. (2 marks)

Question 22**(16 marks)**

A 1.55 kg remote control car is accelerated from rest. The car takes 5.66 s to reach its maximum speed of 14.2 m s^{-1} .

(a) (i) Calculate the acceleration of the car. (2 marks)

(ii) Determine the force needed to accelerate the car. (2 marks)

(b) Calculate the gain in the car's kinetic energy. Express your answer with appropriate units. (3 marks)

- (c) Assuming no energy losses, calculate the power of the motor when the car is travelling at its maximum speed. Express your answer with appropriate units. (3 marks)

- (d) Despite the force due to the power output of the motor, the car is unable to go any faster. Explain why the remote control car has a top speed when considering its motion in real life. Use one of Newton's Laws to help your explanation. (3 marks)

- (e) The front of the remote control car has a soft, bendable bumper that is attached to the car with a spring. Apply your understanding of change of momentum to explain how the bumper keeps the car from becoming too badly damaged in a collision. (3 marks)

Question 23

(11 marks)

An experiment completed by Jess and Ben in a Stage 2 Physics class is outlined below.

7.87×10^{-2} kg of an unknown metal is heated in an oven until it reaches 95.0°C . It is then transferred to an insulated copper calorimeter containing 7.50×10^{-2} kg water, both of which are at 18.0°C . The copper that makes up the calorimeter has a mass of 4.43×10^{-2} kg with a specific heat capacity of $440 \text{ J kg}^{-1} \text{ K}^{-1}$. The final temperature of 25.0°C is recorded after equilibrium is achieved.

Jess and Ben are able to derive formulae for the changes that are occurring:

$$\text{Metal's energy change} = m_{\text{metal}} \times c_{\text{metal}} \times (T_f - T_i)_{\text{metal}}$$

$$\text{Calorimeter's energy change} = \left[m_{\text{copper}} \times c_{\text{copper}} \times (T_f - T_i)_{\text{copper}} \right] + \left[m_{\text{water}} \times c_{\text{water}} \times (T_f - T_i)_{\text{water}} \right]$$

They know that energy should be conserved, so the magnitude of the two energy changes should be equal.

- (a) Calculate the specific heat capacity of the unknown metal given the above information. (4 marks)

(b) It is difficult to determine the exact value of the specific heat due to laboratory conditions.

- (i) What is one possible source of error that Jess and Ben might have encountered? (1 mark)

- (ii) How would you expect the students' calculated specific heat capacity value to compare to the known value? (2 marks)

- (iii) What could they do to improve their results? (1 mark)

(c) Jess and Ben had to write a conclusion to their experiment, but found it difficult to distinguish between heat and temperature. Write a simple definition for the terms 'heat' and 'temperature' as they would apply to the experiment above.

- (i) Heat (2 marks)

- (ii) Temperature (1 mark)

Section Three: Comprehension

10% (17 Marks)

This section has **one (1)** question. You must answer this question. Write your answer in the spaces provided.

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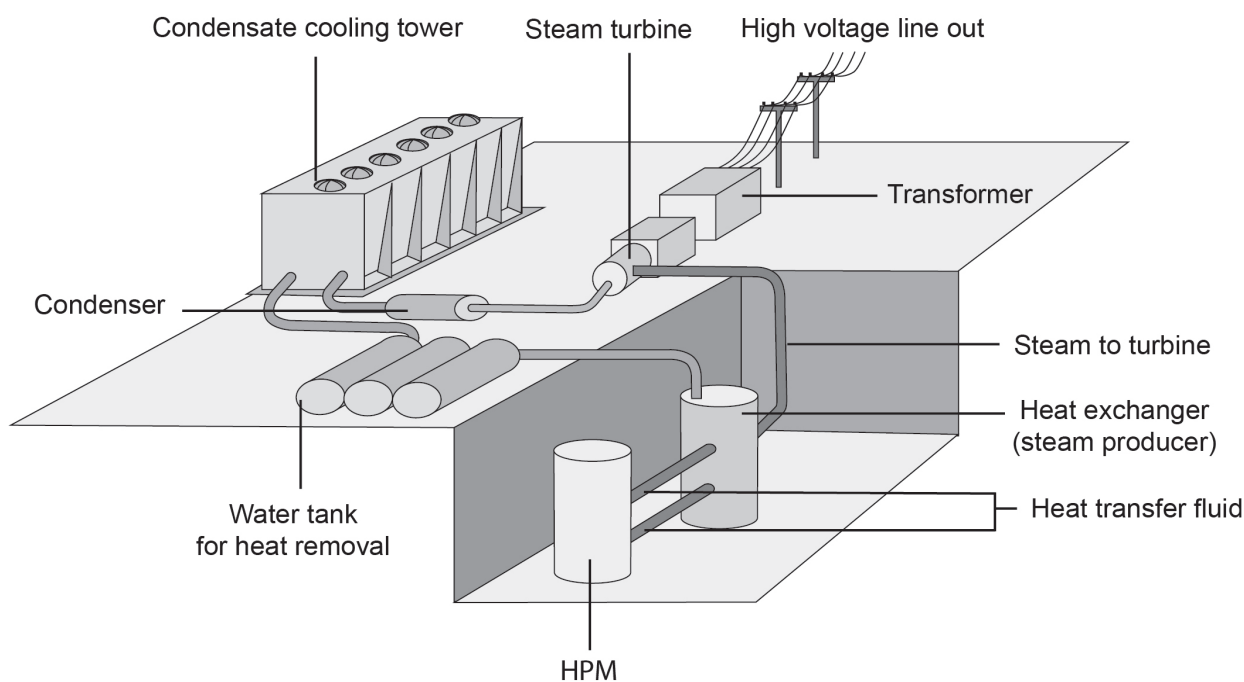
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Suggested working time: 20 minutes.

Question 24

Hyperion Power Generation Inc. is developing a next generation nuclear reactor called the Hyperion Power Module (HPM). It is a liquid metal-cooled small modular nuclear reactor that employs boron carbide (B_4C) control rods to control the reaction. Fuelled by uranium nitride, the HPM has an output of 70.0 megawatts (MW) of thermal power, which is converted to 25.0 MW of electrical power for a 10-year lifetime, without refuelling.



See next page

The HPM produces 25.0 MW of electricity that could power remote mining, oil and gas operations, large government complexes or remote and island communities. A smaller nuclear reactor such as this is more appropriately sized for smaller generation requirements, and can directly replace existing diesel-fuelled generators. It requires no upgrading of existing small electricity distribution systems. The HPM was designed so that it will provide safe and reliable power that is always available and emits no greenhouse gases. It will be manufactured in a factory, transported to the installation site completely sealed, and after its useful life has been reached, replaced with an entirely new power module.

- (a) The article states that 70.0 MW of thermal (heat) power are converted into 25.0 MW of electrical power. Explain why there is a difference in the quantities. (2 marks)

- (b) The last paragraph states that the nuclear reactor can be used to replace an existing diesel-fuelled generator. In the diagram on page 30, circle the part of the power station that would be the same as for a diesel-fuelled power station. (1 mark)

- (c) The safety control system includes three independent shut down systems in the core: a shutdown rod system composed of six boron carbide rods, a control rod system comprising 12 boron carbide rods and a reserve shutdown system consisting of a central cavity into which boron carbide balls may be inserted. Each of the three systems can independently take the core to long-term cold shutdown. The rod shutdown and the ball shutdown systems perform this safety function automatically and instantaneously when triggered.

Using the concept of neutron-induced fission, explain how the boron carbide is able to 'shut down' the nuclear reactor. (2 marks)

- (d) Following the tsunami and nuclear incident near Fukushima, Japan in March 2011, there has been worldwide concern for the prevention of similar events in the future. The technology has improved over time and some differences between the Fukushima and Hyperion Power Module nuclear reactors are listed in the table below.

	Fukushima reactor	HPM reactor	Improvement
Design era	1950s–1960s	2000s–2010s	Incorporates 50 years of reactor operating experience
Coolant	Water (boils at 100°C)	Pb-Bi Metal (boils at >1700°C)	Coolant is highly unlikely to ever evaporate
Containment	Aboveground structure	Underground silo	Better environmental isolation
Decay heat removal	Active, electric power needed	Passively safe for more than 14 days	Less susceptible in accident scenarios
Size	Large	Small	Simplified earthquake resistance

- (i) Choose one point from the table or the article that illustrates an advantage of using the HPM instead of a Fukushima-type of nuclear power station and explain why you believe it to be an advantage. (2 marks)

- (ii) Although advances in nuclear reactors have made nuclear power safer and more easily managed, there are still some problems. Using information from the article, the table or your course work, choose one point that illustrates a problem of nuclear power stations and explain why you believe it to be a problem. (2 marks)

- (e) Most power stations are able to provide electrical power at a potential difference of 1.10×10^4 V to local areas through a substation. Given that an HPM produces 25.0 MW of electricity for a local area at 1.10×10^4 V, calculate the amount of current that is available. (3 marks)
- (f) Over its 10.0 year operational life, the plant will convert the binding energy of nuclides through fission processes to produce a continuous output of 70.0 MW of thermal power.
- (i) Calculate how many joules of energy the reactor would deliver over the 10.0 year period. (2 marks)
- (ii) Using the calculated energy in (i) and the formula $E=mc^2$, determine the decrease in the mass of the nuclear reactor during this time. If you were unable to determine a value for energy in (i) use 2.00×10^{17} J. (3 marks)

Additional working space

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[illegible]

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[illegible]

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[illegible]

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[illegible]

ACKNOWLEDGEMENTS

Section One

- Questions 9–10** Gaddis, L. (2009, June 26). *Lava sample collection* [Photograph]. United States Department of Interior, United States Geological Survey. Retrieved April 4, 2012, from <http://hvo.wr.usgs.gov/multimedia/archive/>.
- Gaddis, L. (2009, June 26). *Collected molten lava* [Photograph]. United States Department of Interior, United States Geological Survey. Retrieved April 4, 2012, from <http://hvo.wr.usgs.gov/multimedia/archive/>.
- Question 11** Photograph of ice skaters used by courtesy of the examining panel.
- Question 16** Photograph of plastic pen used by courtesy of the examining panel.

Section Two

- Question 18** Photographs of batteries and cell used by courtesy of the examining panel.

Section Three

- Question 24** Data source: Harrell, E. (2011, February 28). Nuclear batteries. *Time*. Retrieved January 9, 2012, from www.time.com/time/magazine/article/0,9171,2050039,00.html.
- Data source: Van Loon, J., & Morales, A. (2010). *Fridge-sized nuclear reactors to tap \$135 billion power market*. Retrieved January 9, 2012, from www.bloomberg.com/news/2010-05-16/hyperion-s-fridge-sized-nuclearreactors-to-tap-135-billion-power-market.html.
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